



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	ADVANCED ENERGY MATERIALS				
Type of Course	ELECTIVE IN MAJOR				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Completion of an introductory organic chemistry or material science course is required. 2. Prior knowledge of semiconductor physics, particularly related to energy bands and semiconductor devices, is necessary.				
Course Summary	This course covers essential components and technologies for sustainable energy production. Starting with global energy needs and renewable sources, it explores electrode processes and solar cell materials. Students learn about photovoltaic principles, fuel cell types, and energy storage methods like batteries and supercapacitors. The course also discusses emerging technologies for renewable energy conversion, such as solar thermal and water splitting. Through this, students gain a solid understanding of materials vital for sustainable energy solutions and future energy technologies.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Comprehend the basic principles of energy harvesting and conversion technologies and the significance of materials in these processes.	U	F	Instructor-created exams / Quiz /Assignment
CO2	Assess the performance of different energy harvesting materials and devices, considering factors like efficiency and environmental impact	An	C	Class test /Assignment /Quiz

CO3	Engineer and refine systems for energy production using various materials	Ap	C	Assignment/ Class test
CO4	Grasp how energy storage technologies operate, including batteries and supercapacitors	U	C	Assignments /Seminar presentation
CO5	Evaluate the strengths and weaknesses of energy storage materials and technologies, focusing on factors such as lifespan and energy capacity	An	C	Assignments /Seminar presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Energy Requirement		8	14
	1	World energy requirement. Need for sustainable energy sources.	1	
	2	Sustainable Sun's energy. Current status of renewable energy sources.	2	
	3	Overview of electrode processes. Reversible cells and irreversible cell reactions.	3	
	4	Primary and Secondary cells.	2	
II	Materials for Energy Harvesting		16	26
	5	Solar Cells: Solar spectra, Semiconductors as Solar cell materials. P-N junction diode – Energy band diagram.	2	
	6	Principles of photovoltaic energy conversion – generation of photovoltage. I-V curves of solar cells.	3	
	7	Types of photovoltaic Cells. First generation solar cell materials - Single and polycrystalline Silicon, Amorphous silicon.	3	
	8	Second generation solar cell materials: CdSe, CdTe, Copper Indium Gallium Selenide.	3	
	9	Third generation solar cell materials - Quantum Dots, Organic materials, Dyes.	3	

	10	Types of Organic solar cells - Dye-sensitized solar cell (DSSC) and Polymer solar cells – General overview only.	2	
III	Materials for Energy Conversion		12	20
	11	Fuel Cells: General overview of fuel cell technology.	1	
	12	Types of fuel cells - Alkaline, solid oxide, proton exchange membrane, and Direct methanol. Materials for electrodes, electrolytes in Fuel Cells.	4	
	13	Working principles of H ₂ -O ₂ fuel cell.	1	
	14	Hydrogen economy. Hydrogen generation and storage; limitations. Recent progress in fuel cells.	3	
	15	Piezoelectric and Pyroelectric materials – Energy conversion mechanism with examples.	2	
	16	Thermo-electrics materials – Energy conversion mechanism with examples.	1	
IV	Materials for Energy Storage		12	20
	17	Different types of batteries	1	
	18	Electrode materials, electrolyte and cell reactions of Dry/Alkaline cell and Mercury cell battery. Discharge characteristics, Energy density.	2	
	19	Electrode materials, electrolyte and cell reactions of Lead-acid battery and Ni-Cd battery, Discharge characteristics, Energy density.	2	
	20	Electrode materials, electrolyte and cell reactions of Ni-Hydrogen battery and Lithium-ion/Lithium-polymer battery. Discharge characteristics, Energy density.	2	
	21	Supercapacitors- Types of Electrochemical Supercapacitors.	2	
	22	Electrode and electrolyte interfaces and their capacitances, Charge-Discharge characteristics, Energy/power density.	3	
V	Renewable energy conversion methods		12	20
	Open Ended	Solar thermal technologies, Water splitting and photocatalysis, Energy-related environmental aspects: CO ₂ capture, utilization, and conversion; recovery and recycling of energy materials.	12	

References:

1. Chetan Singh Solanki, Solar photovoltaics : fundamentals, technologies and applications - Third ed. - New Delhi PHI Learning Private Limited, 2011.

2. P.H. Rieger, Electrochemistry, Prentice-Hall, 1987.
3. B.K. Hodge, Alternate Energy Systems and Applications, John Wiley & sons, 2010.
4. A.J. Bard, L.R. Faulkner, Electrochemical Methods, Fundamentals and Application. Wiley, 2001.
5. C. Brabec, Organic Photovoltaics, Wiley-VCH, 2008.
6. Bandarenka, A.S. (2022). Energy Materials: A Short Introduction to Functional Materials for Energy Conversion and Storage (1st ed.). CRC Press.
7. V. Hacker, S. Mitsushima (Eds.), Fuel Cells and Hydrogen: From Fundamentals to Applied Research, Elsevier, 2018
8. K.E. Aifantis, S.A. Hackney, and R. V. Kumar (Ed.) High Energy Density Lithium Batteries Materials, Engineering, Applications, WILEY-VCH Verlag GmbH & Co. KGaA, 2010
9. Yu, A., Chabot, V., & Zhang, J. (2013). Electrochemical Supercapacitors for Energy Storage and Delivery: Fundamentals and Applications (1st ed.). CRC Press

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1					1	3		1		2		1
CO 2	2				1	2	3		1		3		2
CO 3				1	2	1	3		2				3
CO 4					3	1	3		2		3		3
CO 5				2		3	3		2		2		3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓